Hardware in the Loop Simulation of Direct Synthesis based Two Degree of Freedom PID Control of DC-DC Boost Converter using Real Time Digital Simulation in FPGA

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Abstract— **The second order DC-DC boost converter is a non-minimum phase system as its control-to-output (c2o) transfer function contains a zero in the right half plane. This non-minimum phase behaviour poses several challenges to control design like initial inverse transient response to step changes, smaller bandwidth etc. Conventional One Degree of Freedom (1DOF) controllers, when applied to Pulse Width Modulated (PWM) boost type DC-DC converter suffer from the compromise that has to be made between the set-point response and disturbance rejection. The 2DOF-PID control scheme provides an extra degree of freedom to overcome above mentioned shortcoming. In 2DOF-PID, the feedback controller is designed through Direct Synthesis (DS) approach, where the controller transfer function is derived using Maclaurin series. Further, the rapid prototyping tools such as FPGA based Xilinx System Generator (XSG) facilitate quick realization of real time controllers before they are finally used for field testing. The results obtained by implementing controller in real time FPGA environment, using Hardware in the Loop (HIL) techniques, show the robustness of the presented control scheme to regulate the output voltage of boost DC-DC converter system.**

Keywords— PWM Boost type dc-dc converter, Direct Synthesis (DS), 2DOF-PID,Field Programmable Gate Array (FPGA), Hardware in Loop (HIL) simulation, Real Time Digital Simulator (RTDS), Voltage regulation, Xilinx System Generator (XSG)

I. INTRODUCTION

For the last two decades, the excessive use of computers and telecommunication equipment has resulted in high demand for the switch mode power supplies. The fast growing market of clean and pollution free renewable energy domains like fuel cell, solar photovoltaic and wind energy rely on effective and efficient conversion of electrical power. Hence, the use of power electronic converters (e.g. DC-DC and DC-AC) has become very common. In general, DC-DC converters are non-linear and time varying in nature and are subjected to varying line voltages and wide range of uncertain load changes. Due to these conditions, converter's performance deteriorates significantly. Inspite of a lot of research on high performance control techniques for DC-DC power converters, there exists a need for effective and flexible controllers which can perform tight regulation under unpredictable line and load disturbances. A boost type DC-DC converter in particular has an RHP zero that depends on the load, which is highly uncertain [1]. Hence, from the control design perspective, this makes the control of boost converter a challenging task for the researchers and imposes limitation on the achievable performance.

The transfer function approach, employed to mathematically model DC-DC power converters, is used to design most of the linear controllers. Methods available for modelling these DC-DC switching power converters are: (i) State space averaging method; (ii) Circuit averaging and (iii) Averaged switch modelling [2]. The state space averaging method is very commonly used technique for the modelling and design of controller for switched mode dc-dc power converters.

Due to the presence of RHP zero in c2o transfer function of boost converter, the single feedback closed loop crossover frequency goes well below that of RHP zero which results in sluggish response of the system. In addition to this, under feedback control of switched mode dc-dc boost type PWM converter, its c2o transfer function varies non-linearly with duty cycle [3, 4]. These inherent properties of a boost converter impose challenging constraints in output voltage regulation.

A specific level of performance of the controlled system must be ensured irrespective of certain limited variations occurring in the process dynamics. These performance criterions of a controller are judged by three performance indices: [5] (1) Tracking of desired reference signal; (2) Bandwidth of the controlled process and (3) Robustness with process variations and external disturbances.

Can a single loop feedback controller achieve all these afore mentioned performance criterions? The answer is no, because of:

(1) Algebraic Limitations: To achieve proper tracking, both the sensitivity (S) and complementary sensitivity (T) function should be maintained to low value.

In conventional 1DOF control scheme as shown in Fig. 1, if e', r', d', y' and n' denote the error, reference signal, disturbance, output, and noise respectively then $e' = r' - y' = n' \times T + (r' - d') \times S$ .Since $|S| + |T| = 1$, both S and T can't be reduced simultaneously [6]. Thus, to ensure minimal tracking error, feedback control law needs to be developed such that the magnitudes of T and S should be small in the respective intervals where frequency components of n' and e' are significantly large [5].

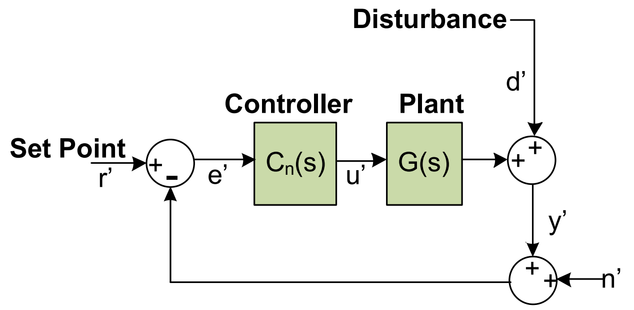

Fig. 1. Conventional 1DOF control scheme

(2) Bandwidth limitation: For achieving higher bandwidth, S should be lower, so that T is close to 1 upto a large frequency. But this selection of large frequency is limited by the hardware sampling rate and water bed effect problem. Also, the controller gain can't be rolled off very quickly within certain interval of frequency range, because it causes instability in the system. Moreover, due to non-minimum phase nature of the system, the achievable bandwidth becomes very small [7]. Because of this, the attenuation of the noise by using high roll of rates of the controller gains is not possible.
(3) Robustness in the presence of variation of plant parameters and disturbances to plant variations and disturbances.

So, for achieving the above desired performance indices in the presence of aforementioned limitations, S and T are made independent in the frequency range of setpoint changes and disturbance rejection. Now, the pre-filter $F(s)$ or any feed-forward controller is designed to create a steep decrease in the gains at higher frequencies so as to reduce T thereby speeding up the setpoint response [8]. This way appreciable output voltage regulation and tracking for disturbances and set-point changes are achieved independently through feedback control $C(s)$ and pre-filter $F(s)$ respectively. Upto certain high frequency, usually bandwidth of compensated system (ω_{bw}), the feedback controller dominates, and after that pre-filter or feed-forward controller dominates upto gain crossover frequency of T (ω_T). This increase in the degrees of freedom in 2DOF-PID control design allows the designer to completely attain the performance criterion.

Recently, an attempt has been made in the formulation of 2DOF control scheme [9] to improve the steady state and dynamic performances of non-minimum phase system such as DC-DC boost converter.

Conventional 1DOF control schemes are not able to attain the desired voltage regulation under source and load perturbations and to track the changes made in the set point simultaneously. To overcome the limitations reported about 1DOF control schemes it is recommended to use 2DOF control scheme.

In the presented work, a 2DOF-PID controller, has been designed and tuned using direct synthesis approach [10, 11, 12], and is employed as a control technique for the regulation of output voltage of the boost converter. The proposed scheme has the following merits-

(1) Design of pre-filter for set-point response and feedback PID control for the line and load disturbances are decoupled from each other which improves the output voltage regulation, disturbance rejection and gives superior transient and steady state response.

(2) The feedback PID controller in this scheme has only one tuning parameter (λ) in contrast to the conventional PID control with three tuning parameters (K_p, K_i, K_d).

(3) Desired setpoint response is achieved by tuning the pre-filter through tuning parameter γ.

The controller implemented on real time FPGA hardware package interacts with computer system power circuit models giving authenticity to the simulation and ensuring zero error in the last stage of field testing of the proposed scheme [13]. This is shown in Fig. 2.

The remaining paper is organised as follows: The second section concisely explains about the boost converter considered in the present work and its control design procedure based on Direct Synthesis 2DOF-PID method. Section III gives a detailed description and stepwise procedure of XSG based real time hardware in loop simulation. Section IV is dedicated to presentation of real time HIL simulation results obtained with controller implemented on FPGA. Section V concludes this paper.

II. DESIGN RELATIONS FOR DS BASED 2DOF-PID CONTROL

DC-DC boost converter considered in the presented work is operated in CCM and is shown in Fig. 1.

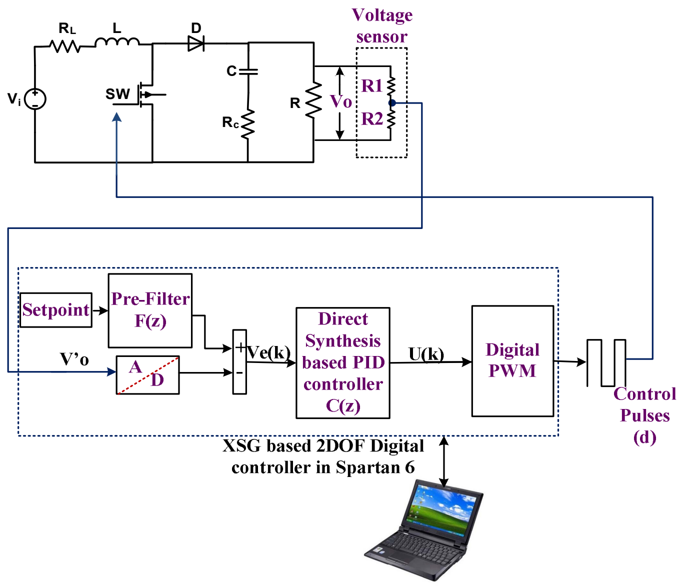

Fig. 2 HIL simulation of 2DOF-PID control of a boost converter using FPGA based Spartan 6 development board.

Table 1. The parameters selected for the DC-DC boost converter

Parameter	Numerical values
V_{in}(V)	50
R(Ω)	56.26
L(mH)	2.2
C(μF)	100
D	0.34
R_L(Ω)	0.1
R_c(Ω)	0.2
V_o(V)	75

State space averaging method [2] is used to derive following c2o transfer function of boost converter and the relevant parameters are tabulated in Table 1

$$\frac{\hat{v}_o}{\hat{d}}(s) = \frac{111.3338(1+2\times10^{-5}\,s)(1-9.0448\times10^{-5}\,s)}{(5.0105\times10^{-7}s^2+0.00015692s+1)} \quad (1)$$

Most of the existing control design methods suffer from a major shortcoming of not having decoupled sets of tuning parameters which can help the designer to independently tune the setpoint response and disturbance rejection. Recently reported control scheme [9] was an attempt to overcome the existing shortcoming by designing a 2 DOF control taking maximum sensitivity in to consideration as a parameter of robustness. But this attempt remains untouched to the output voltage regulation under source disturbance and commanded voltage change. Control structure shown in Fig. 3 is an alternative approach to developing design relations for decoupled tuning of pre-filter and feedback PID control through γ and λ as their tuning parameter respectively, using the concept of direct synthesis reported in [10, 11].

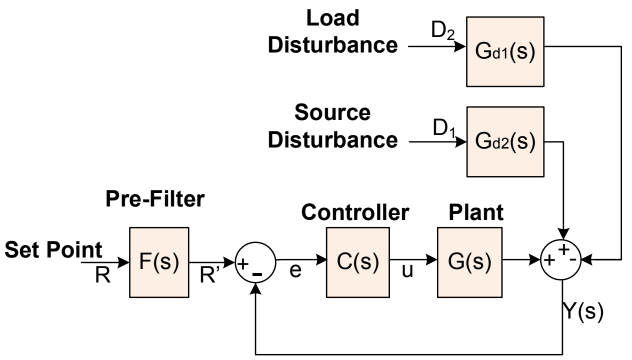

Fig. 3 2DOF-PID control structure

Here, the design procedure consists of two steps: (i) Design of feedback PID control with λ as tuning parameter and (ii) Design of pre-filter with γ as tuning parameter

In the DS approach, the desired output behaviour is specified as a trajectory. The controller is designed based on the plant model, which helps the output to follow this trajectory accurately [12]. This DS approach leads to: (i) Easily designable and tuneable feedback PID controller with only one tuning parameter (λ) and (ii) Better performance under line and load fluctuations along with time varying output reference signal. The DS approach has been incorporated with 2DOF control scheme resulting in better control performance.

The c2o transfer function of boost converter can be factored as:

$$G(s) = G^-(s)G^+(s) \quad (2)$$

where $G^-(s)$ is the invertible and $G^+(s)$ is the non invertible part. For the c2o transfer function written in time constant form, $G^+(0)=1$ gives the least square response [10]. The transfer function between the measured output and pre-filter output (R') can be written as:

$$T_{R'Y}(s) = \frac{G(s)C(s)}{1+G(s)C(s)} \quad (3)$$

The transfer function $T_{R'Y}(s)$, using (2) can further be written as:

$$T_{R'Y}(s) = \frac{G^+(s)\Psi(s)}{(\lambda s+1)^r} \quad (4)$$

where tuning parameter λ controls the trade-off between robustness and performance. The variable r is chosen such that $T_{R'Y}(s)$ becomes a proper transfer function [11]. $\psi(s)$ is a polynomial defined as:

$$\psi(s) = 1+\sum_{i=1}^{p}\beta_i s^i \quad (5)$$

where p of $\psi(s)$ denotes the number of poles in $G_D(s)$ intended to be cancelled by the poles of the controller. In general, p is of the order not more than 2 and β_i is selected so as to cancel the desired poles in $G_d(s)$. The coefficients β_i are solved by using the relation

$$\left|1-T_{R'Y}(s)\right|_{s=r_1,r_2} = 0 \quad (6)$$

where $r_1\, r_2$ are the distinct poles of the boost converter (1), and $T_{R'Y}(s)$ [11] is

$$T_{R'Y}(s) = \frac{G^+(s)(\beta_2 s^2+\beta_1 s+1)}{(\lambda s+1)^4} \quad (7)$$

The value of β_i obtained using (1), (6) and (7) are $\beta_1 = 4.9454\times10^{-4}, \beta_2 = 1.1042\times10^{-7}$

from (3), $C(s)$ can also be expressed as:

$$C(s) = T_{R'Y}(s)\left[G(s)(1-T_{R'Y}(s))\right]^{-1} \quad (8)$$

using (1), (7) and (8), the controller $C(s)$ can be expressed as

$$C(s) = \frac{f(s)}{s} \quad (9)$$

$f(s)$ can be expanded by using Maclaurin series

$$C(s) = \frac{1}{s}\left(f(0)+sf'(0)+s^2 f''(0)\right) \quad (10$$

Comparing with standard PID controller expression [11] tuning parameters thus obtained, are listed in Table 2. Final expression for feedback PID control thus becomes:

$$C(s) = \frac{1.77\times10^{-7}s^2+1.233\times10^{-6}s+0.00224}{6.577\times10^{-8}s^2+0.0003749s} \quad (11)$$

The expression for pre-filter in 2DOF-PID [11] can be written as:

$$F(s) = \frac{(\gamma\beta_1 s+1)}{(\beta_2 s^2+\beta_1 s+1)} \quad (12)$$

where $0\le\gamma\le1$ and the calculated pre-filter, $F(s)$ is as below:

$$F(s) = \frac{0.0001978s+1}{1.104\times10^{-7}s^2+0.0004945s+1} \quad (13)$$

Table 2. The parameters of 2DOF-PID control scheme through DS approach

S. No.	Controller gain	Numerical Value
1	Proportional	5.9755
2	Integral	4.7182×10^{-6}
3	Derivative	0.0022

III. DESIGN RELATIONS FOR DS BASED 2DOF-PID CONTROL

XSG, is a toolset of MATLAB/simulink and Xilinx IP core blocks, which can generate VHDL code using HDL co-simulation and this does not require expertise in VHDL/verilog code writing.

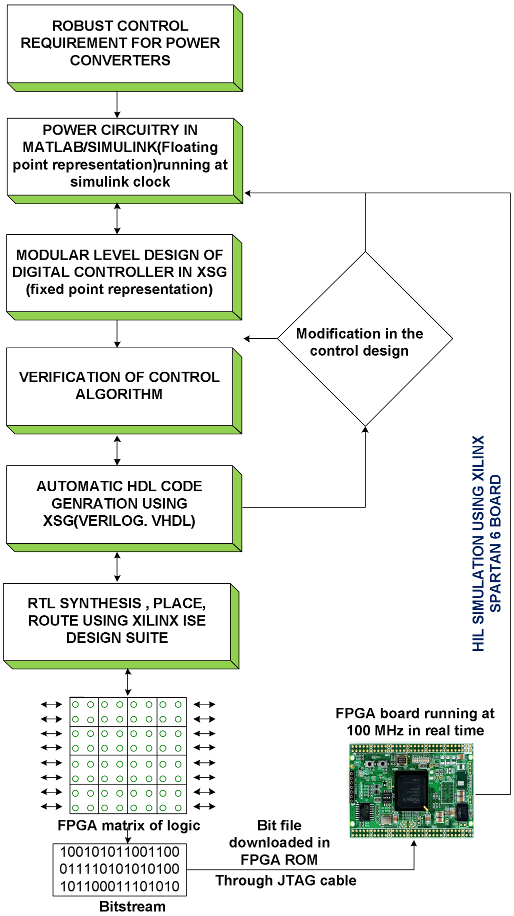

Fig. 4 Flow chart of HIL simulation using XSG and FPGA board.

The FPGA controller designed for a converter control in the HIL simulation can be used with the actual system after testing. This helps in rapid prototyping of the low cost and easily programmable FPGA controller for power electronics converters applications. The closed loop 2DOF-PID controller based PWM DC-DC boost converter is simulated in real time using HIL technique in FPGA. The stepwise description for implementing HIL using XSG [14, 15] is given in Fig. 4. When the bit stream is loaded into ROM of FPGA, all Xilinx blocks are replaced by single HIL block, which represents the real time physical controller implemented in FPGA and interacting with virtual power circuitry through Joint Test Action Group (JTAG) cable.

IV. DESIGN RELATIONS FOR DS BASED 2DOF-PID CONTROL

This section gives the detailed analysis of the presented work. The chief merit of the proposed 2DOF-PID control implementation in real time Spartan 6 FPGA board, using HIL simulation through XSG, is the verification of system's dynamic performance in the presence of actual FPGA controller. To examine the suitability and usefulness of the considered 2DOF-PID control, boost converter has been operated under different operating conditions, which are: (A) Varying the output commanded voltage in step manner; (B) Step decremental variation in the input excitation and (C) Step up alteration in the load.

A. Step variation in commanded output voltage

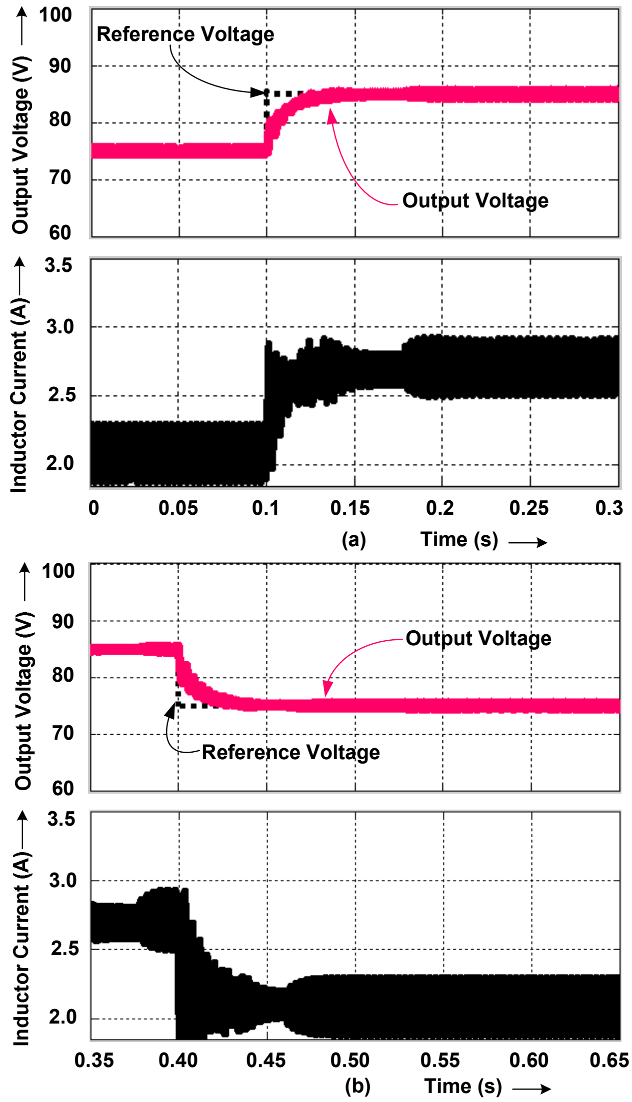

Fig. 5 HIL Simulation results obtained through actual controller implementation on Spartan board, for output voltage and inductor current under: (a) Step-up variation in commanded output voltage; (b) Step down variation in commanded output voltage.

The converter is operated at a switching frequency of 40 kHz. System was operating in steady state and at t = 0.1 sec step-up change of 10 V is introduced which is shown in Fig. 5 (a). The notable points are: (i) HIL simulation results thus obtained for considerable changes show the ability of 2DOF-PID controller in tracking the new output voltage reference, without any appreciable oscillations and overshoot, in about 20 ms. (ii) Furthermore, the closed loop response to a step down change in reference voltage is shown in Fig. 5 (b). It can be inferred from Fig. 5 (b) that the controller stabilizes and regulates the desired output voltage in a short time span of 23 ms without any undershoot.

The corresponding change in inductor current, to maintain the desired regulated output voltage, is also shown in Figs. 5 (a) & (b).

B. *Step decrement in input volatge*

System was already operating in steady state corresponding to V_{ref} = 75V when input voltage is changed from 50V to 46V, in a step manner, at t = 0.75 sec. Fig. 6 and the corresponding enlarged view depicts that the controller regulates the desired output voltage and the new steady state operating point is reached in a time of about 17 ms

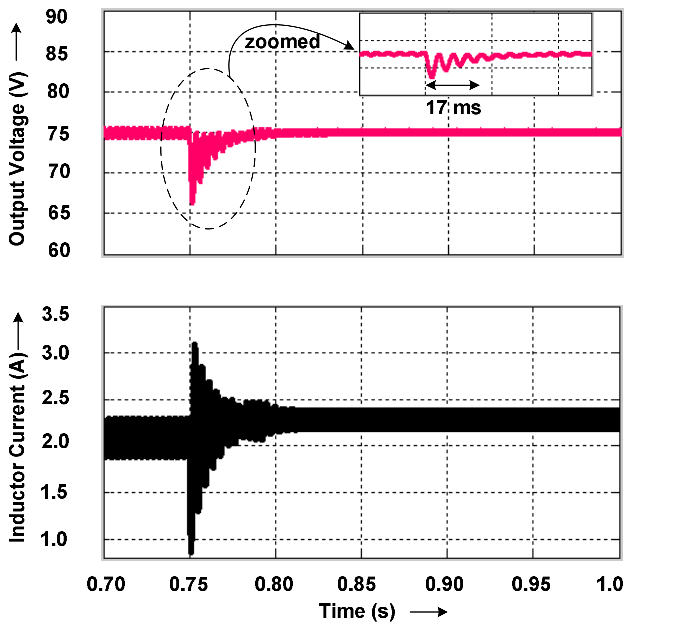

Fig. 6 Transient response for output voltage and inductor current of a boost converter under step down source voltage perturbation.

C. *Step Change in Load*

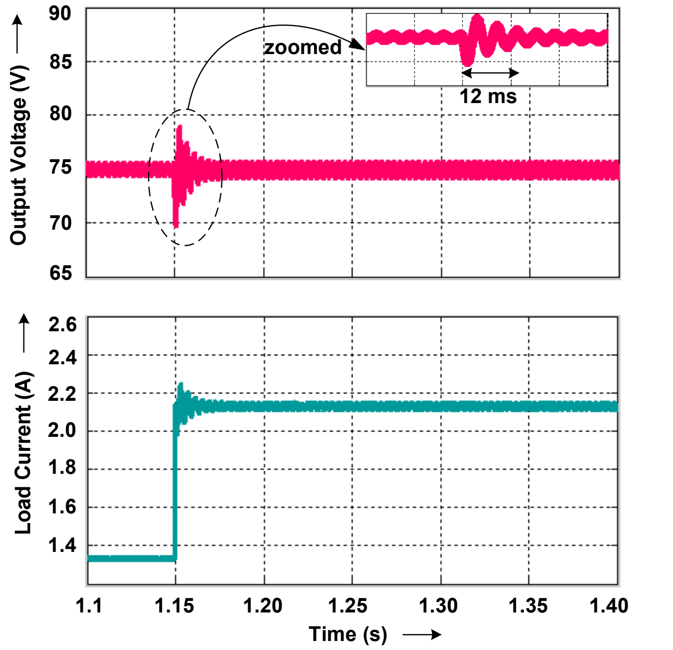

Fig. 7 HIL simulation results for load current and output voltage under 60 % step up change in load current from its nominal point.

The performance of the controller is also examined for step change in the load. As shown in Fig. 7(a), step down change in resistance from R = 56.26 to 35.16 Ω, occurs at 1.15 sec, which correspond to about 60% change in the load current (the input voltage is maintained at 50V and output voltage reference is 75V). It is clearly depicted in Fig. 7. The corresponding enlarged view show that after the converter settles to new operating point, the controller stabilizes the output voltage back to 75 V in an appreciably short span of almost 12 ms.

V. Conclusion

A DS based 2DOF-PID control scheme has been proposed and is implemented in FPGA, to regulate the desired voltage output of a DC-DC boost PWM converter operated in CCM mode. Rapid prototyping tool XSG has been used for real time realization control scheme. The effectiveness and performance of 2DOF-PID control scheme, for boost converter of the proposed work, has been tested in real time using Spartan FPGA board. The outcome in HIL simulation environment shows the flexibility and robustness of proposed scheme in achieving the desired output voltage tracking and regulation in real conditions of line-load perturbations and setpoint changes with appreciably fast dynamics and stable steady state response.

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